

# Shreyas Srinivas Bikumalla

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## EDUCATION

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| <b>College of Engineering, Michigan State University, East Lansing, MI</b><br><i>Master of Science, Computer Science</i> | Aug 2023 – May 2025<br>GPA: 3.80 |
| <b>Vishwakarma Institute of Information Technology, Pune, India</b><br><i>Bachelor of Science, Computer Science</i>      | Aug 2019 – May 2023<br>GPA: 3.90 |

## WORK EXPERIENCE

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| <b>Software Engineer   Python, GitHub, ML, AWS, REST APIs, Jenkins</b><br><b>Michigan State University, East Lansing, MI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Jul 2025 – Present  |
| <ul style="list-style-type: none"><li>Optimized machine learning models to achieve 90% accuracy in locust brain odor compound detection across multiple time windows</li><li>Developed Python pipelines processing 10-second spatio-temporal datasets to extract features for neural spike classification.</li><li>Mentored professors and 5+ lab colleagues in Python programming and machine learning for collaborative neuroscience research.</li></ul>                                                                                                                                                                                                                                                                                                                                                                  |                     |
| <b>Graduate Teaching Assistant   Python, C++, Flask, Docker, REST APIs, SQL, JavaScript</b><br><b>Michigan State University, East Lansing, MI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Aug 2024 – May 2025 |
| <ul style="list-style-type: none"><li>Directed 2 large-scale undergraduate courses in OOP, databases, and cloud-based app development for 200+ students.</li><li>Mentored students in C++, SQL, microservices, Docker, Flask and OOP design, improving project outcomes by 20%.</li><li>Led 10+ Web App Development Workshops, reducing student errors by 15% and accelerating project delivery by 35%.</li><li>Optimized 50+ student projects through JavaScript, cloud deployment, and API integration support, improving efficiency by 25%.</li></ul>                                                                                                                                                                                                                                                                    |                     |
| <b>Data Analyst   Python, MySQL, Snowflake, Jenkins, ETL Pipelines, GitLab, CI/CD</b><br><b>Colgate-Palmolive, Mumbai, India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Jan 2023 – May 2023 |
| <ul style="list-style-type: none"><li>Coordinated with global business and product teams to design data analytics solutions that support strategic initiatives.</li><li>Automated 20+ financial workflows and improved dashboard/DAX reporting accuracy by 25% across 20+ regions.</li><li>Engineered scalable data visualization dashboards, improving product performance tracking efficiency by 30%.</li><li>Architected a multicurrency reporting system in MySQL and SAP, reducing manual processing time by 40%.</li><li>Deployed a Snowflake to MySQL ETL pipeline, cutting data refresh cycles by 60% through GitLab-driven CI/CD automation.</li><li>Maximized data reliability and stakeholder reporting efficiency by 35% through scalable pipeline integration and optimized communication workflows.</li></ul> |                     |
| <b>Software Engineer   Python, JavaScript, TypeScript, AWS, Go, React, Redux, PostgreSQL, REST APIs</b><br><b>Innobytes, Pune, India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sep 2022 – Dec 2022 |
| <ul style="list-style-type: none"><li>Designed and developed a full-stack financial web platform to improve credit access for micro-entrepreneurs.</li><li>Directed back-end integration and API development, streamlining approval workflows by 50% and ensuring scalability.</li><li>Adapted the entrepreneur-lender transaction times by 60% by building a responsive platform with JavaScript, React, Redux, CSS, PostgreSQL and Python on AWS using EC2, S3, RDS.</li><li>Tuned back-end processing speed by 20% by integrating RESTful APIs</li><li>Enhanced prediction accuracy by 30% by upgrading ML-based approval models with scikit-learn, PyTorch</li></ul>                                                                                                                                                    |                     |

## CERTIFICATIONS

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|--------------------------------------------------|---------------|----------|
| Google Data Analytics Professional               | <b>Google</b> | May 2025 |
| Cisco Certified Network Associate (CCNA) - 1,2,3 | <b>Cisco</b>  | Dec 2022 |

## PROJECTS

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| <b>GCN for Node Classification   Python, Docker, Jenkins, TensorFlow, AWS, RAG, GitHub, LangChain</b>                                                                                                                                                                                     | Jan 2025 – May 2025 |
| <ul style="list-style-type: none"><li>Implemented a PyTorch GCN model that achieved 93.7% node classification accuracy, a 10% improvement over baseline</li><li>Increased validation accuracy by 10% and reduced overfitting by 25% through scikit-learn hyperparameter tuning.</li></ul> |                     |
| <b>Deep Learning model for Speech Enhancement   Python, CUDA, React Native, PyTest, LangGraph</b>                                                                                                                                                                                         | Jul 2024 – Dec 2024 |
| <ul style="list-style-type: none"><li>Led development of a Conv-TasNet deep learning speech enhancement model, improving PESQ by 50%+</li><li>Achieved a 30% STOI increase by evaluating on large-scale datasets and visualizing KPIs for real-time performance monitoring.</li></ul>     |                     |

## TECHNICAL SKILLS

- Programming:** Python, C++, JavaScript, TypeScript, SQL, R, HTML5, CSS3, XML, OOP, Data Structures & Algorithms
- Frontend & Backend:** React, Redux, Angular, Vue, Flask, REST APIs, Node.js, CI/CD, Postman, FastAPI, GraphQL
- Machine Learning & Data:** TensorFlow, Keras, PyTorch, RAG, pandas, NumPy, Tableau, LangChain, LangGraph, Fine-tuning
- Cloud & DevOps:** AWS, S3, Lambda, EC2, Azure, Docker, Kubernetes, Git, GitHub, Jenkins, GitLab, System Design
- Databases & Other:** MySQL, PostgreSQL, MongoDB, Agile, Snowflake, Jira, SDLC, Linux, PyTest, JUnit, Jest